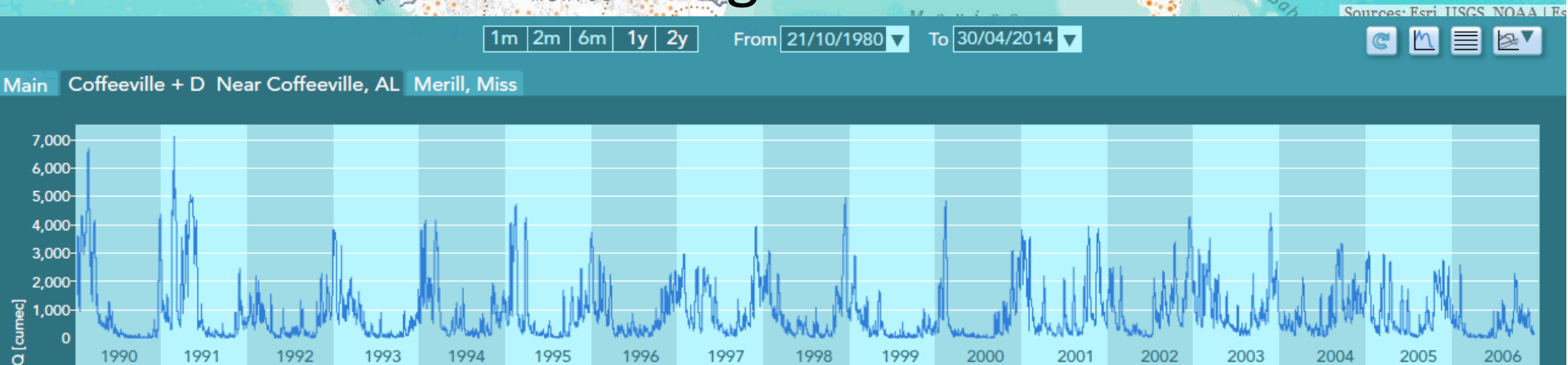


Hydrology Domain Working Group

David R. Maidment

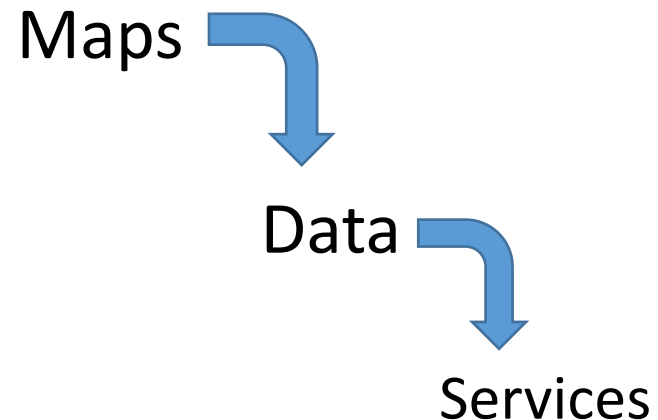
City University of New York
New York City, NY

11 August 2014



Major Transitions in Geospatial Info

- *Paper maps to digital data*
 - National Spatial Data Infrastructure development
 - Started in 1990's
 - Took more than a decade to complete
- *Digital data to web services*
 - Started several years ago
 - Will take years to complete



Geospatial Services

Goal:

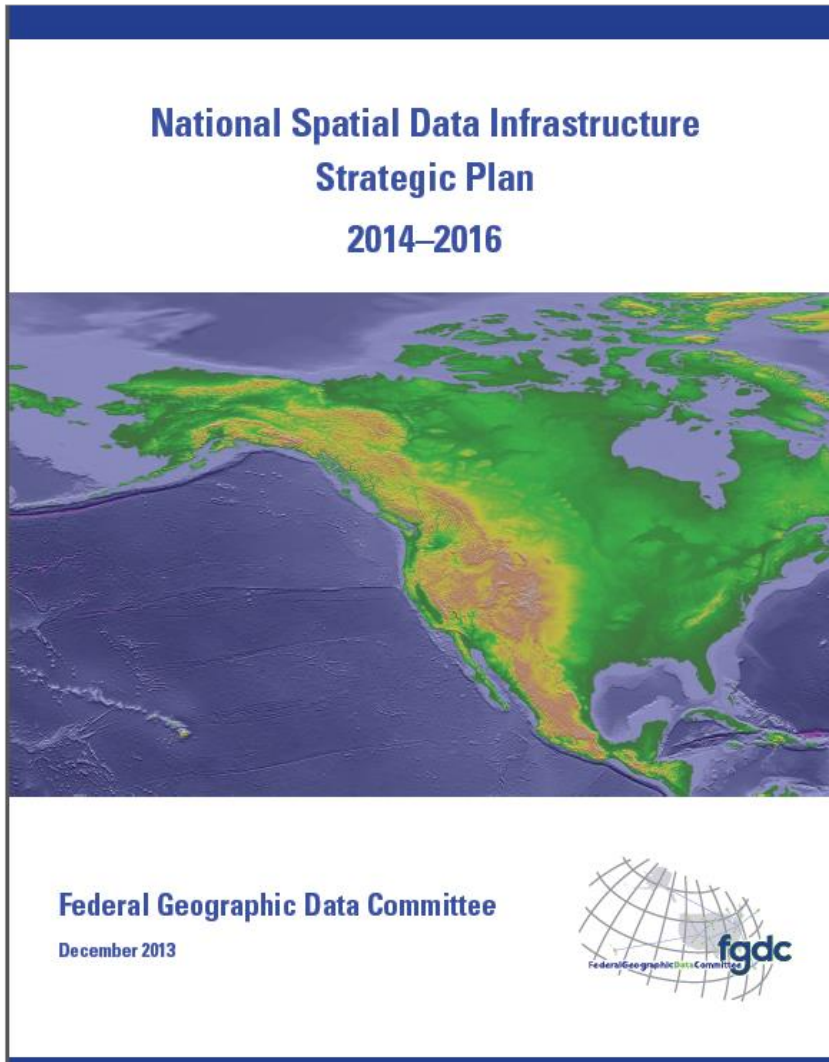
Calculate average available water storage for a set of contributing areas

Key to Geospatial Services: Standardized & Machine-Readable Formats

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6	08173000	6	884402	795961800	20.384517
7	08173900	7	124844	1123601400	18.655235
8	08174600	8	132233	1190102400	17.519698
9	08175000	9	157671	1419043500	17.337202
10	08177700	10	61329	55196100	9.802769
11	08178000	11	55929	50336100	16.753598
12	08178565	12	209875	188887500	18.716674
13	08178700	13	391345	352210500	8.977314
14	08178800	14	143295	128965500	18.604578
15	08178880	15	941543	847388700	5.824062
16	08179000	16	420717	378645300	8.710093
17	08179100	17	161998	145798200	6.803158

- Stream Gages
- Digital Elevation Model
- Watershed Delineation Tool
- Soil Available Moisture Map
- Soil Statistics by Watershed

National Spatial Data Infrastructure (NSDI)

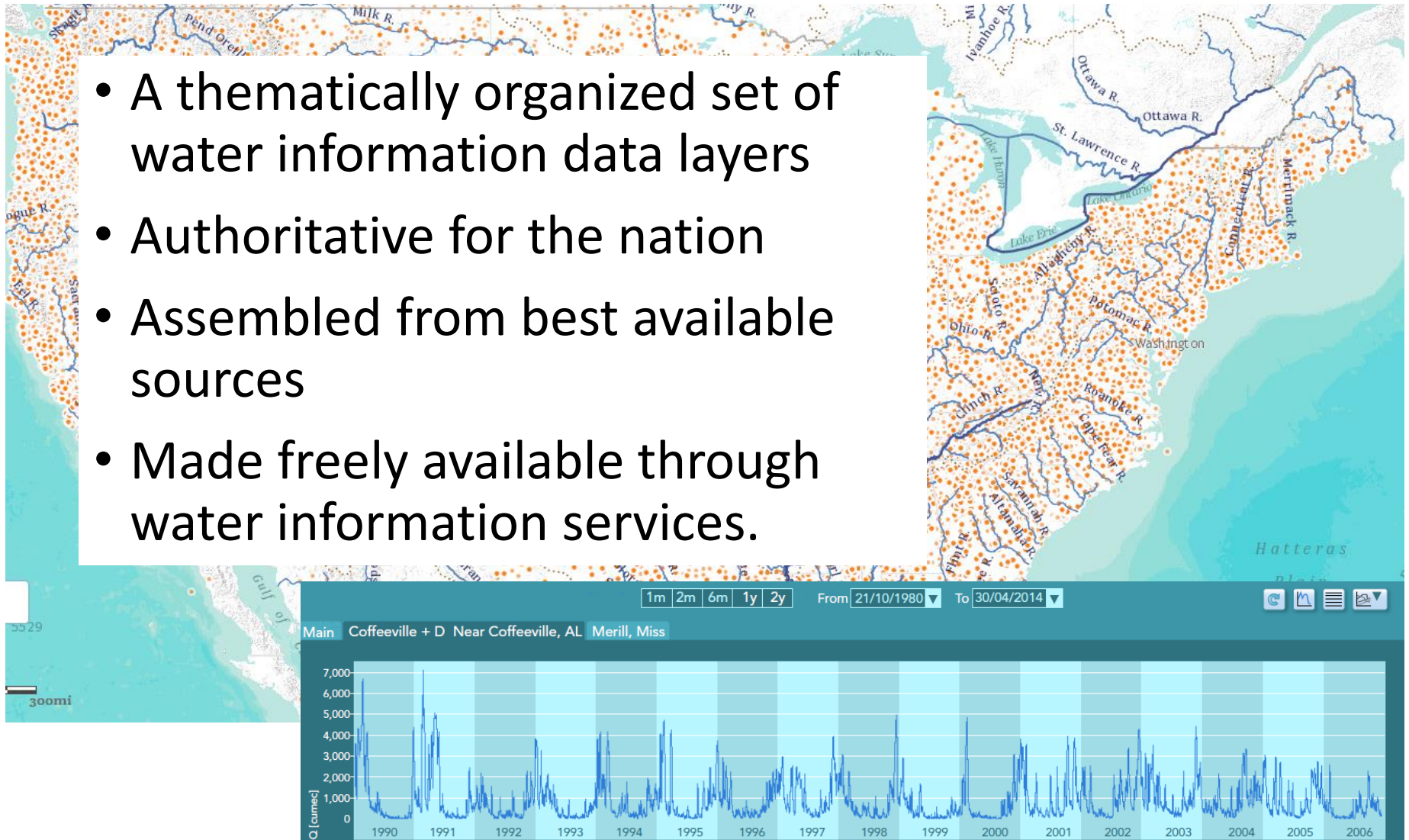


Desired Future State of NSDI

- Create **network** of resources and services
- Facilitate **discovery**, access and application of resources
- Leverage shared **standard**-based services
- Develop core set of information layers that interface with **nonspatial data**
- Use **real-time** data feeds and sensor webs

National Water Data Infrastructure (NWDI)

- A thematically organized set of water information data layers
- Authoritative for the nation
- Assembled from best available sources
- Made freely available through water information services.

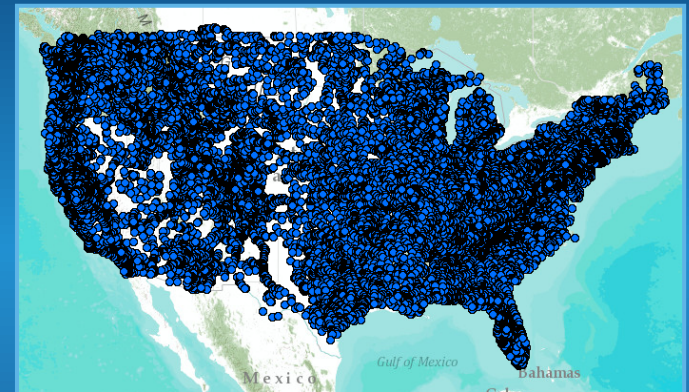


WaterML Web Services – CUAHSI, USGS, OGC, WMO

Water time series data on the internet



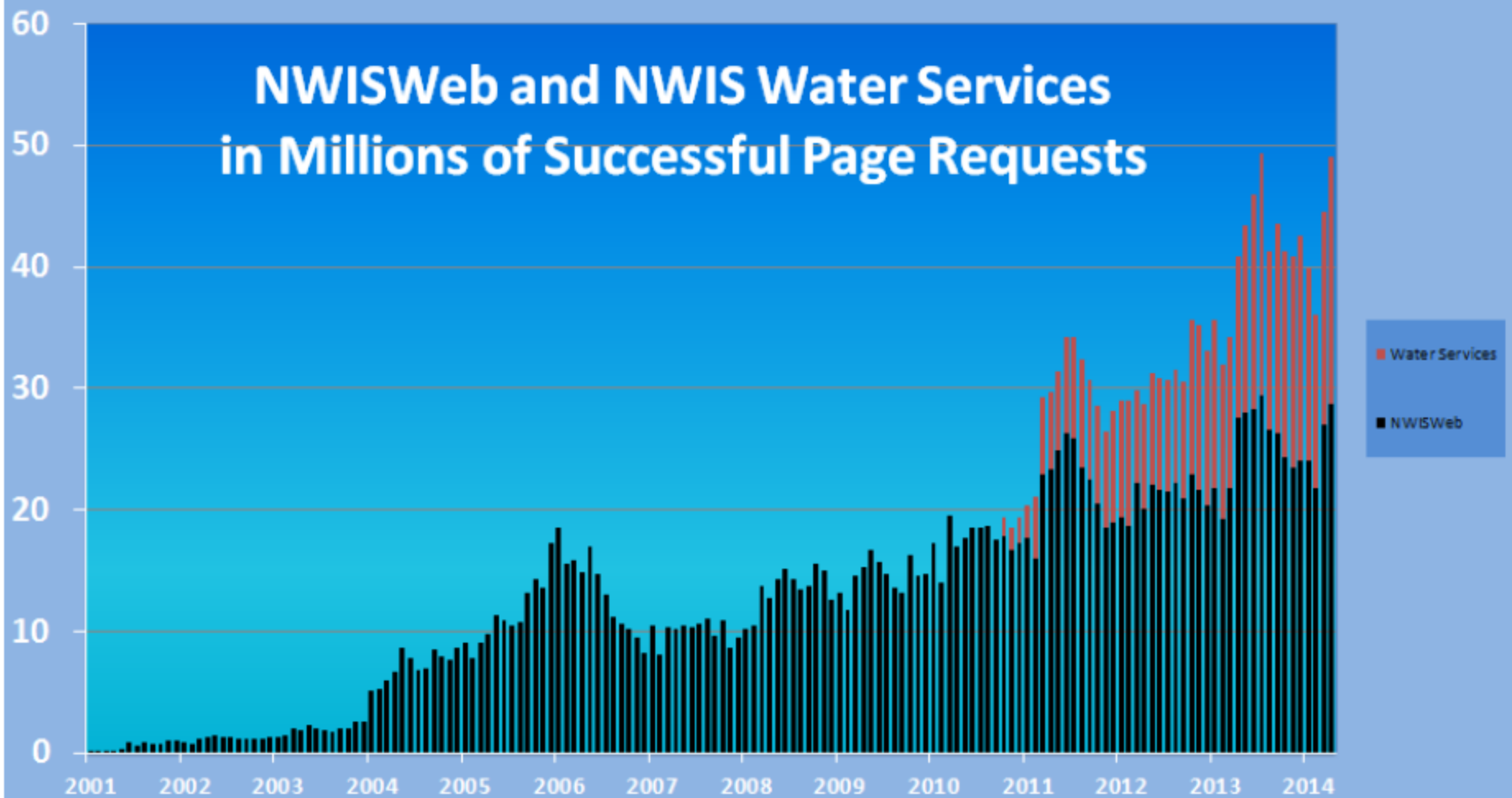
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24/7/365 service
For daily and real-time data

... Operational water web services system for the United States

NWISWeb and NWIS Water Services in Millions of Successful Page Requests



OGC/WMO Hydrology Domain Working Group

4-Year International Effort – WaterML

A time series for one variable at one location

WaterML2

Hydrology Domain Working Group formed
OGC at WMO Commission for Hydrology

November 2009

*MEMORANDUM OF UNDERSTANDING
BETWEEN
THE WORLD METEOROLOGICAL
ORGANIZATION
AND
THE OPEN GEOSPATIAL CONSORTIUM, INC.*

Technical Meetings Each 3 Months
Four Interoperability Experiments
(Surface water, groundwater, forecasting)
Annual week-long workshops
Involvement by many countries

Acknowledgements: OGC, WMO, GRDC, NWS,
CUAHSI, BoM/CSIRO, USGS, GSC, Kisters,

2008

2009

2010

2011

2012



Sponsored by



Short WMO IPET-MDRD report on WaterML 2

**90th OGC Technical Committee
Washington, DC**

**Frédéric Guillaud, Météo-France and Jeremy Tandy, UKMO
25 March 2014**

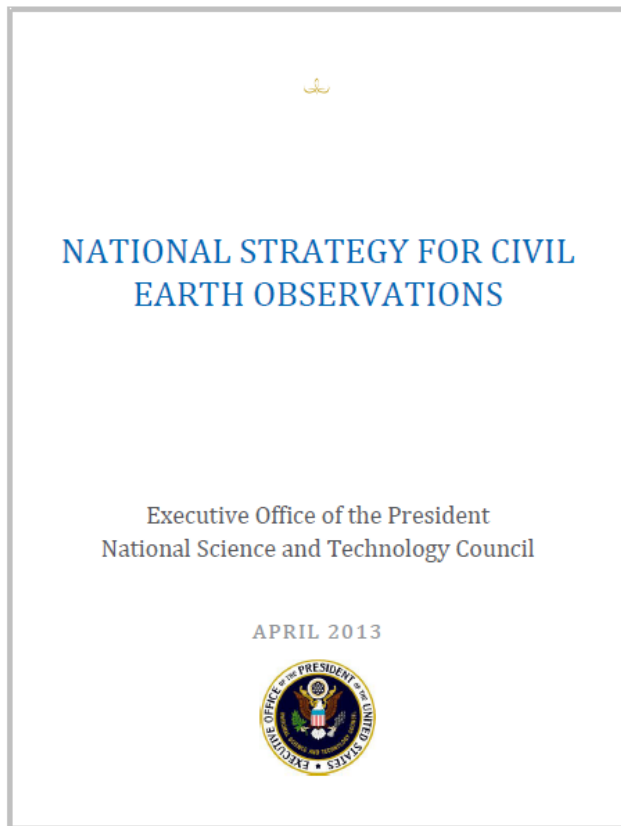
IPET-MDRD Report on WaterML



- **IPET-MDRD : Inter Programme Expert Team on Metadata and Data Representation Development**
- **Meeting held in Silver Spring, hosted by NOAA, last week**
- **There is nothing hydrology-specific in the WaterML2 Part 1 specification.**
- **Inspire IR specifies WaterML Time series for Atmospheric (ACMF) and Oceanographic (OF) themes**
- **Executive Council had asked CBS to lead adoption of WATERML2 as a WMO standard and also its adoption by ISO**
- **Repackage WaterML 2.0: Part 1 as **TimeSeriesML** in order to be used in a broader context (MetOcean)**

OGC/WMO Hydrology Domain Working Group

- **WaterML2** standard for time series approved in 2012
- **Ratings and Gagings** standardization ongoing now within the Hydrology Domain Working Group (driven by requirements for **IWRSS**)
- **RiverML** initiated at Quebec HDWG meeting in 2013
- **This workshop** in New York in August 2014



Committee on Environmental, Natural Resources, and Sustainability (CENRS)

“WaterML: Water Markup Language (ML) is an informatics initiative of the CENRS Subcommittee on Water Availability and Quality that provides a systematic way to access water information from point observation sites.” (p.21)

WaterML is the first OGC standard endorsed by the Executive Office of the President

President's Climate Data Initiative

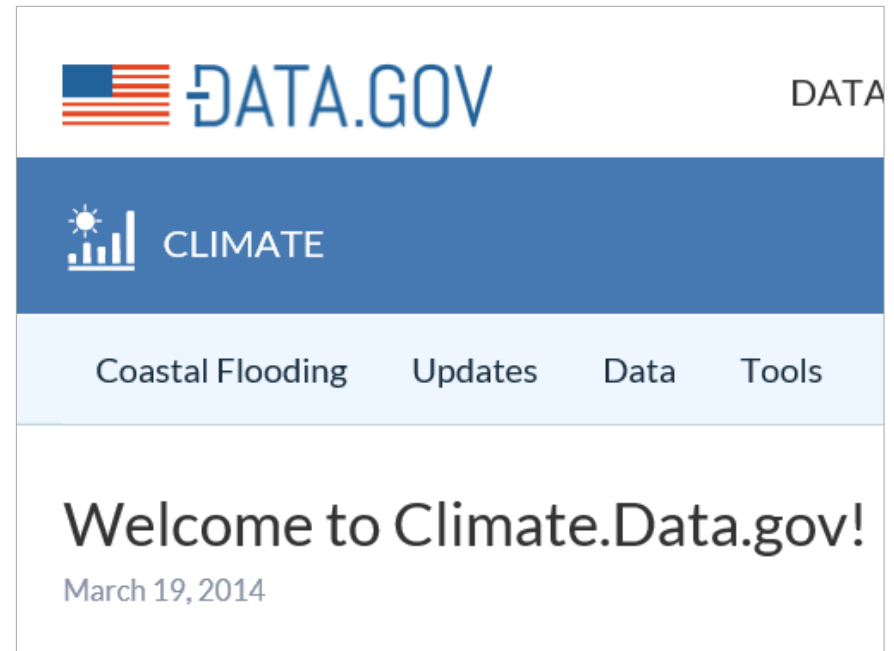


John Podesta, Counsellor to the President, White House, March 19, 2014

<https://www.youtube.com/watch?v=pfe5oRdsCp0&feature=youtu.be&t=7s>

Goals of Climate Data Initiative

- Prepare for **climate change**
- Make **government data** accessible
- Engage **private sector**
 - ESRI, Microsoft, Google, Intel
- Engage **communities**
 - Chesapeake, New Orleans, World Bank, ...



Why not Water.Data.gov?

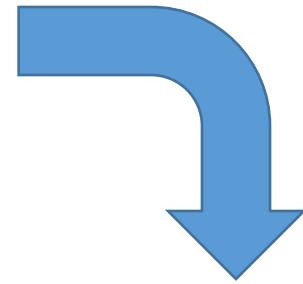
Build an Open Water Data Initiative

Anne Castle, Asst Secretary for Water and Science, Dept of Interior

- “Today we have unprecedented opportunities to use science and technology to create a better understanding of one of our most precious resources – water”
- “I am committed to working with you to do that”



Chair





June 26, 2014

Steering Committee Meeting

Open Water Data Initiative

Kevin T. Gallagher

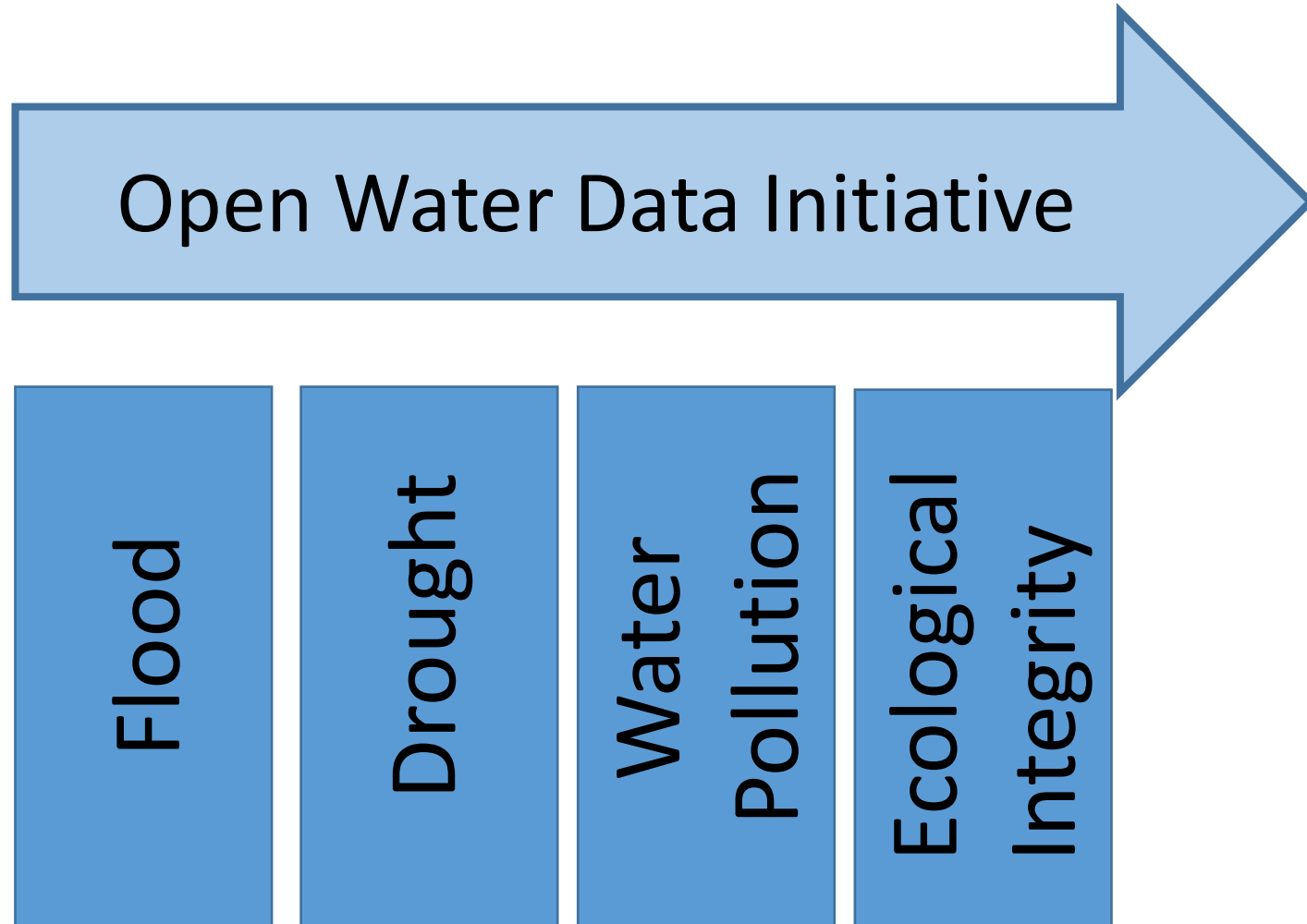
USGS, Associate Director, Core Science Systems

Charge for the FGDC

- ◆ In collaboration with the ACWI and other partners – Advance an Open Water Data Initiative, including:
 - Reviving and populating the joint Subcommittee on Spatial Water Data to design a national open water data infrastructure;
 - Supporting IWRSS consortium members in the scoping and implementation pilot activities;
 - Creating an integrated water data portfolio for specific hydrologic regions or basins;
 - Developing a technical reference architecture that supports the sharing of water data and links observations to geospatial data;
 - Leveraging the Geospatial Platform to make water data more accessible and to support water data community collaboration;



Four Possible Use Cases



ACWI – Advisory Committee on Water Information

WICP Water Information Coordination Program
ACWI Advisory Committee on Water Information

Advisory Committee on Water Information

[▶ About Us](#) ▶ [Members](#) ▶ [Meetings](#) ▶ [Contacts](#) ▶ [Products](#) ▶ [Old Features](#)

Features:
Mark your calendars

9th National Monitoring Conference

April 28 - May 2, 2014

Location:
Cincinnati



Water Information Coordination Program
The WICP ensures collaborative efforts among Federal Agencies to improve water information for decisionmaking about natural resources management and environmental protection.

Advisory Committee on Water Information
The ACWI represents the interests of water-information users and professionals in advising the Federal Government on Federal water-information programs and their effectiveness in meeting the Nation's water-information needs. ([ACWI Fact Sheet](#))

Established in 1991



Overview: National Water Center (NWC)

- Common Operating Picture for Water Resources
- Geo-Intelligence Laboratory
- Science and software studio
- Systems proving ground
- Distance learning

Center opened on 13 May 2014



INTEGRATED WATER RESOURCES SCIENCE AND SERVICES (IWRSS)

SUMMIT TO SEA



Overview: National Water Center (NWC)

◦ Operations Center with Situation Rooms

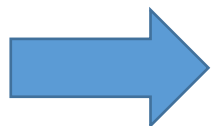
Temporal information

- Establish common operating picture for floods to droughts; begin demonstration of hourly summit-to-sea analyses and forecasts of soil moisture, evapotranspiration, and snow pack; and expand demonstration of Real-Time Dynamic Flood Inundation Mapping portraying the extent, depth, and impacts of flood waters to enhance community resiliency and enable decision makers to mitigate the impacts of floods.

◦ Geo-Intelligence Laboratory

Geospatial information

- Develop, implement and maintain state-of-the science enterprise Geographic Information Systems (GIS) to support NWS operations



National Water Data Infrastructure

Transformative Research (NSF)

Transformative research involves ideas, discoveries, or tools that radically change our understanding of an important existing scientific or engineering concept or educational practice or leads to the creation of a new paradigm or field of science, engineering, or education. Such research challenges current understanding or provides pathways to new frontiers.

http://www.nsf.gov/about/transformative_research/definition.jsp

How to move from *evolutionary* change to *transformative* change?

Email Message to Don Cline, 14 May 2014

- I can see that you have a **very fine facility**, the challenge now is to fill it with people productively working. I gather that you have plans for some federal staff to move to Alabama, but for 2015 at least there will likely be some additional space available. It occurred to me that this might be a nice opportunity for **engagement with the academic community** at the University of Alabama and elsewhere.
- I believe it is possible to have a **real-time flood data, modeling, forecasting and inundation mapping facility running for the continental US in a demonstration mode** at the National Water Center by the end of the summer 2015.

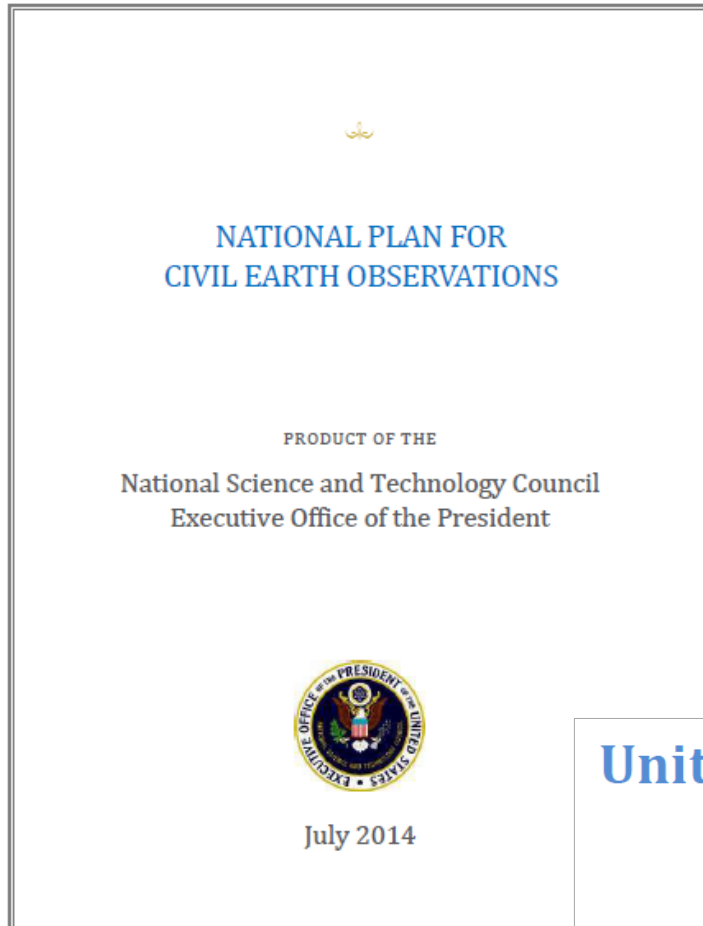
National Flood Interoperability Experiment (NFIE): Connecting national flood forecasting with local flood emergency response

- Will be led by the academic community in collaboration with the IWRSS partners through the National Water Center
- Run from September 2014 to August 2015
 - Preparatory phase to May 2015
 - Summer Institute at the National Water Center to bring things together June to August 2015
- Serves a use case in flooding for the Open Water Data Initiative
- Open to all levels of government, commercial providers, academia

NFIE Academic Centers



National Plan for Civil Earth Observations (Office of Science and Technology Policy, July 2014)



National Science and Technology Council

Chair

John P. Holdren

Assistant to the President for Science and Technology
Director, Office of Science and Technology Policy

NSTC Staff Executive Director

Jayne Morrow

Office of Science and Technology Policy

Committee on the Environment, Natural Resources, and Sustainability

Co-Chairs

Tamara Dickinson

Principal Assistant Director for Environment & Energy
Office of Science and Technology Policy

Glenn Paulson*

Assistant Administrator for Research and Development and Science Advisor
Environmental Protection Agency (A1)

Kathryn Sullivan

Under Secretary for Oceans and Atmosphere
Administrator of the National Oceanic and Atmospheric Administration
Department of Commerce

United States Group on Earth Observations

Chair

Peter Colohan

Office of Science and Technology Policy

United States Group on Earth Observations

Chair

Peter Colohan

Office of Science and Technology Policy



Highest Priority Observations

- Weather and Seasonal Climate Monitoring and Prediction
- Dynamic land-surface monitoring and characterization
- Elevation and Geo-Location
- Water level and flow

3.2.1.1.Tier 1

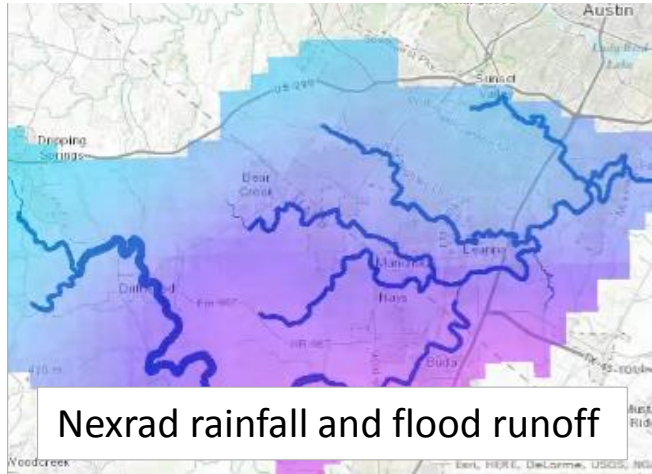
Based on results of the EOA, the following measurement groups support a majority of the societal themes.¹⁷ These measurement groups represent the highest priority measurements in the category of sustained observations for public services. They are listed below in priority order.

1. Weather and seasonal climate monitoring and prediction: Observations in this measurement group characterize phenomena such as precipitation, storms, wind, floods, sea state, drought, wildfires, ice, air quality (including ozone), and weather risks to human health and transportation. They also contribute to short-term climate forecasts. These observations derive from next-generation radar on fixed and mobile platforms; atmospheric sounding from space and airborne platforms, oceanic measurements, and spectral and radiometric imaging of the Earth surface.
2. Dynamic land-surface monitoring and characterization: Observations in this measurement group support food and water security, water availability and quality, fire detection and suppression, human health, forestry, soil characterization (including soil moisture), hazards mapping and response, and natural-resource management. They particularly derive from multispectral and hyperspectral imaging from space and airborne platforms, forest inventory, and soil data collection.
3. Elevation and geo-location: Observations in this measurement group support food and water security, hazard and risk mapping, and natural-resource management. These observations particularly include topography and bathymetry, surface modeling, hydrologic data, and ecosystems-related data as derived from radar and laser sensors on satellite-based, airborne, and terrestrial platforms, as well as positioning, navigation, and timing satellites, such as those used for the Global Positioning System (GPS).
4. Water level and flow: Observations in this measurement group support coastal inundation and inland flooding, water availability, hydropower management, transportation, human health, water equivalent of snow, and tsunami hazard preparedness. They particularly derive from coastal and inland water level and flow measurements, seafloor topography, and ocean topography from satellite altimetry.

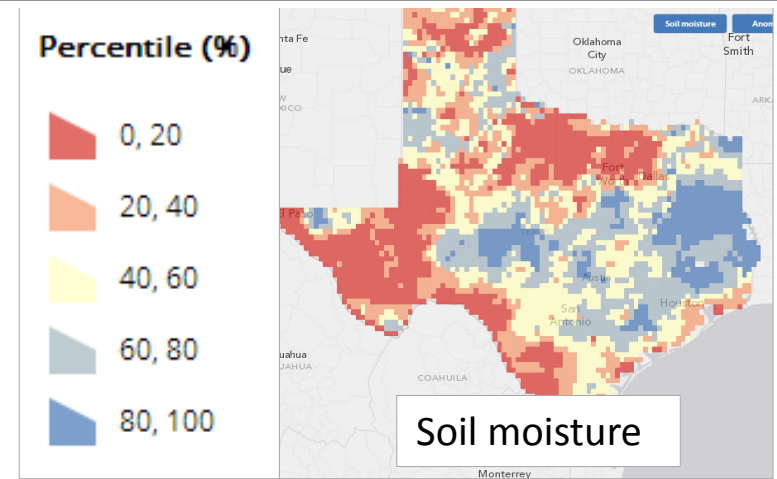
Science goal: Link these four sources of observations to help predict floods and assess their impact on people

Flood observations and modeling

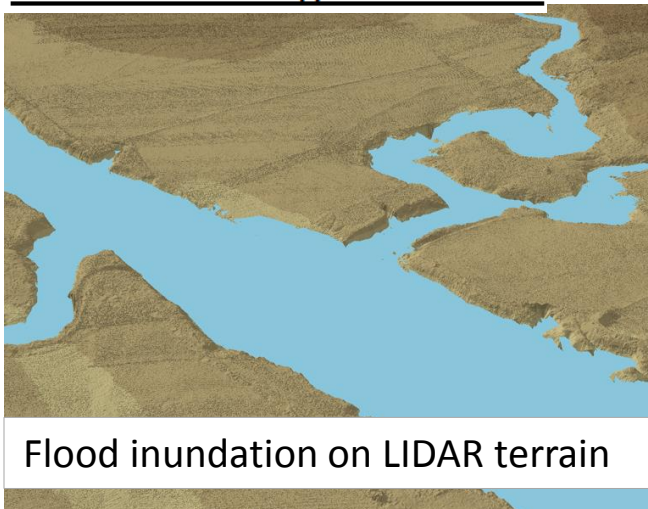
1. Weather and seasonal climate monitoring and prediction



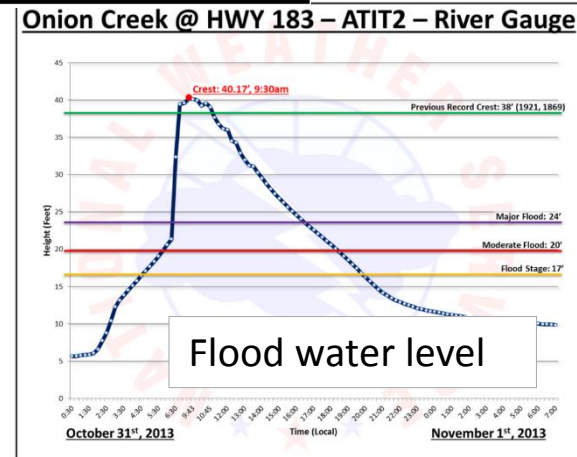
2. Dynamic land-surface monitoring and characterization



3. Elevation and geo-location



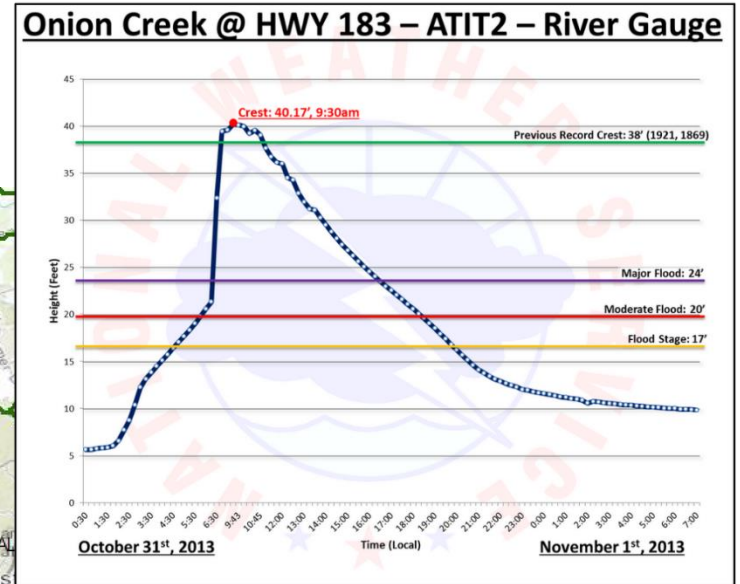
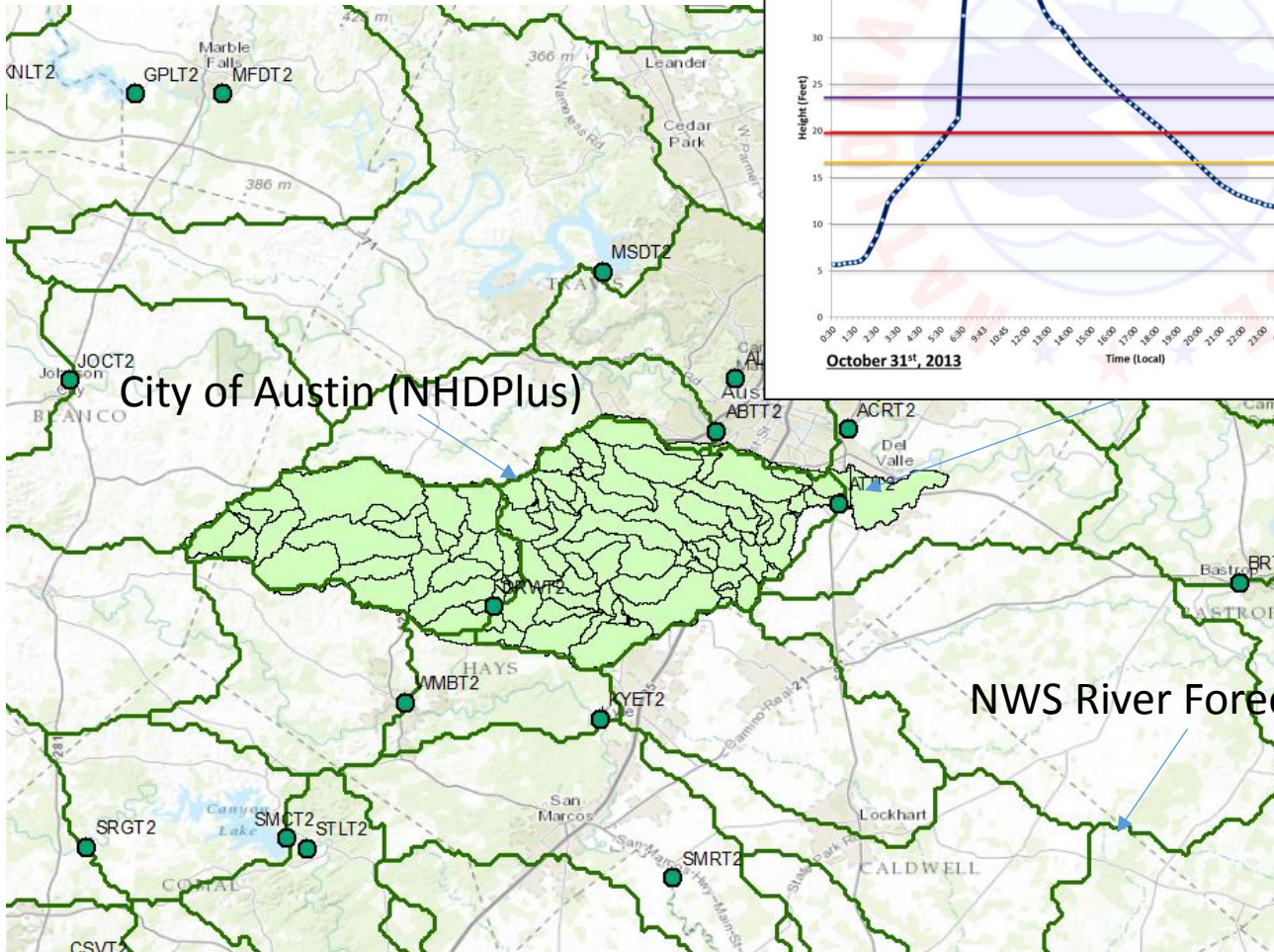
4. Water level and flow



River Forecast Centers

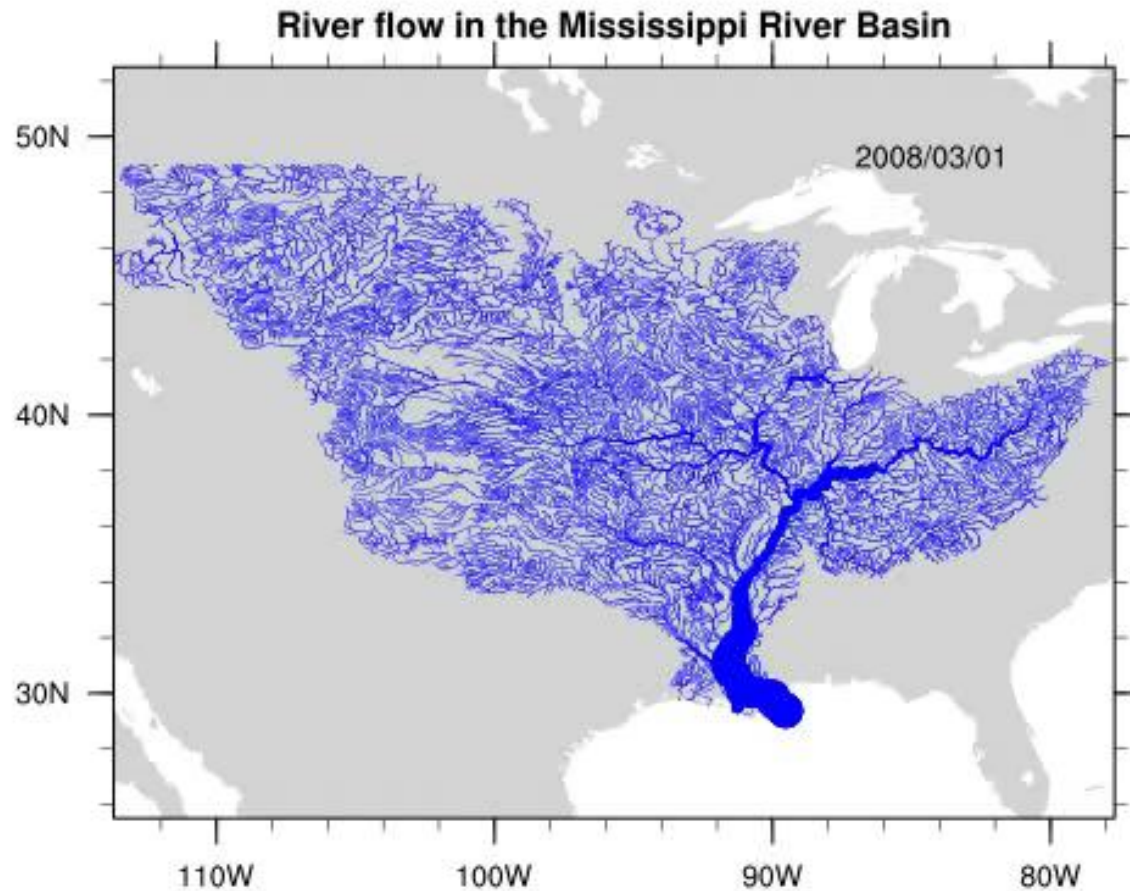


National and Local Flood Modeling



Flow in the Mississippi Basin

RAPID model, 3 hour time steps



<http://www.geo.utexas.edu/scientist/david/rapid.htm>

Tavakoly et al. (201x), in preparation

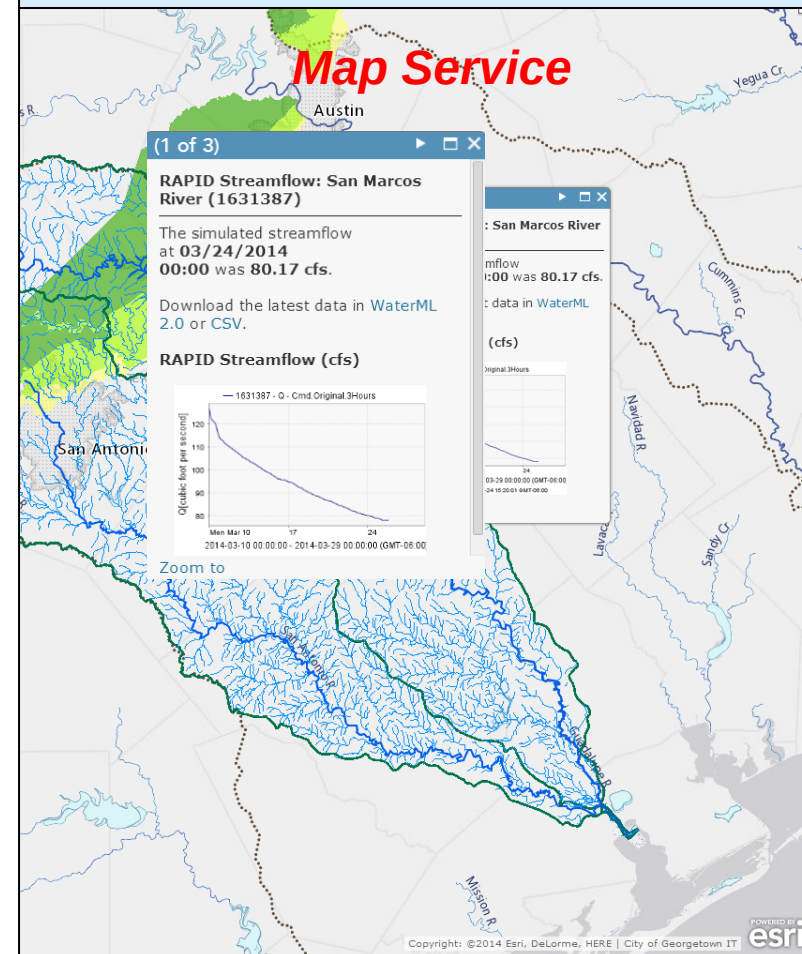
Water Operations Model

RAPID Streamflow Calculations – 5,175 NHDPlus river reaches

Q[cubic foot per second]

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Data Service



Flood hydrology and response

Flood Inundation Map Library

```
graph TD; A[Flood Inundation Map Library] --- B((Flood hydrology and hydraulic data, models, forecasts)); A --- C((Flood emergency response planning and action)); B --- D[Forecast the flood elevation]; C --- E[Determine and plan for flood impact];
```

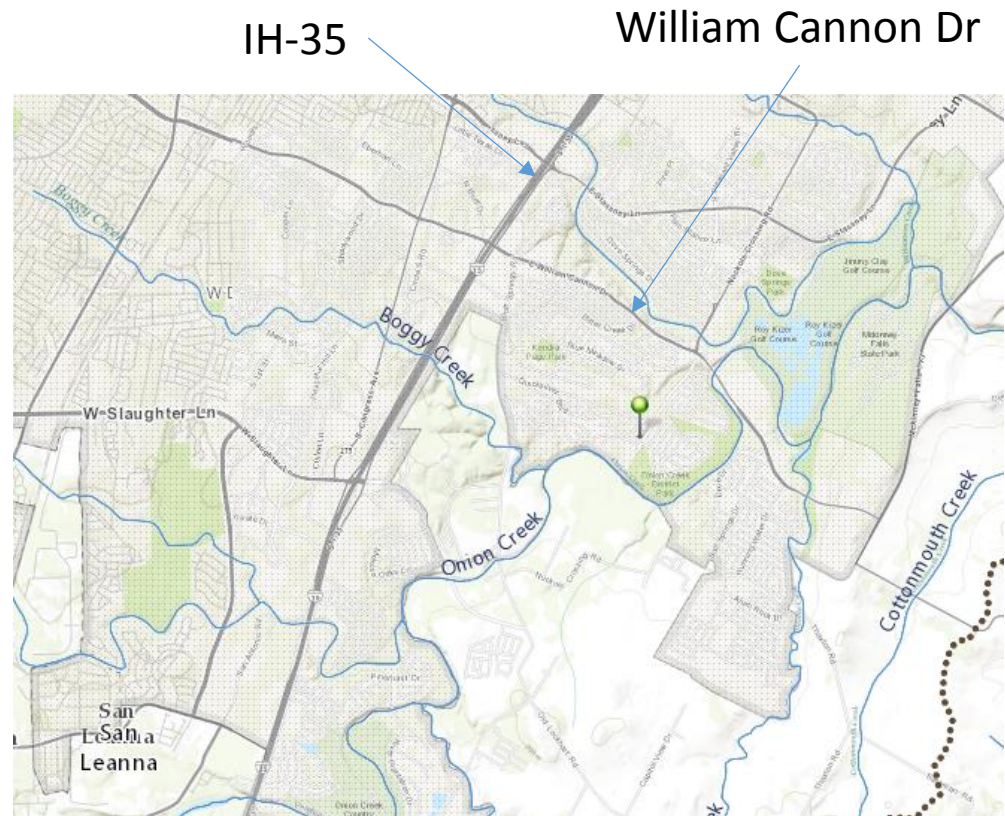
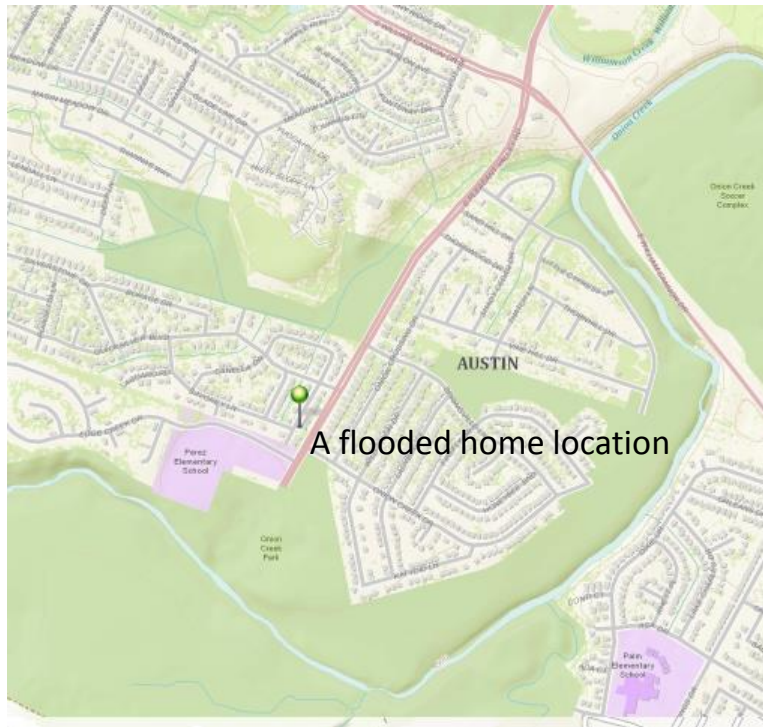
Flood
hydrology and
hydraulic data,
models,
forecasts

Forecast the flood elevation

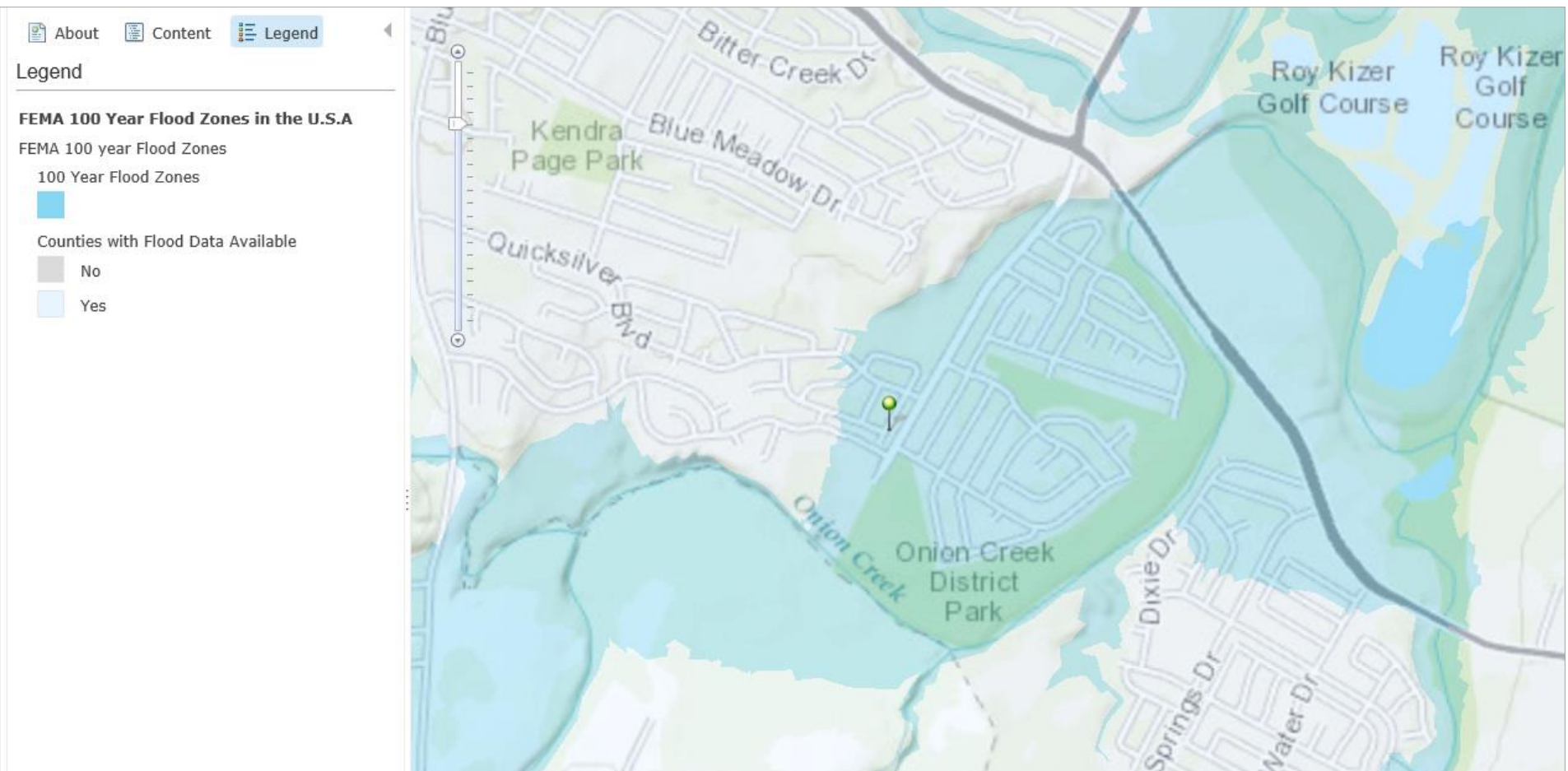
Flood
emergency
response
planning and
action

Determine and plan for flood impact

Halloween Flood, Onion Creek, Austin, Texas, October 2013

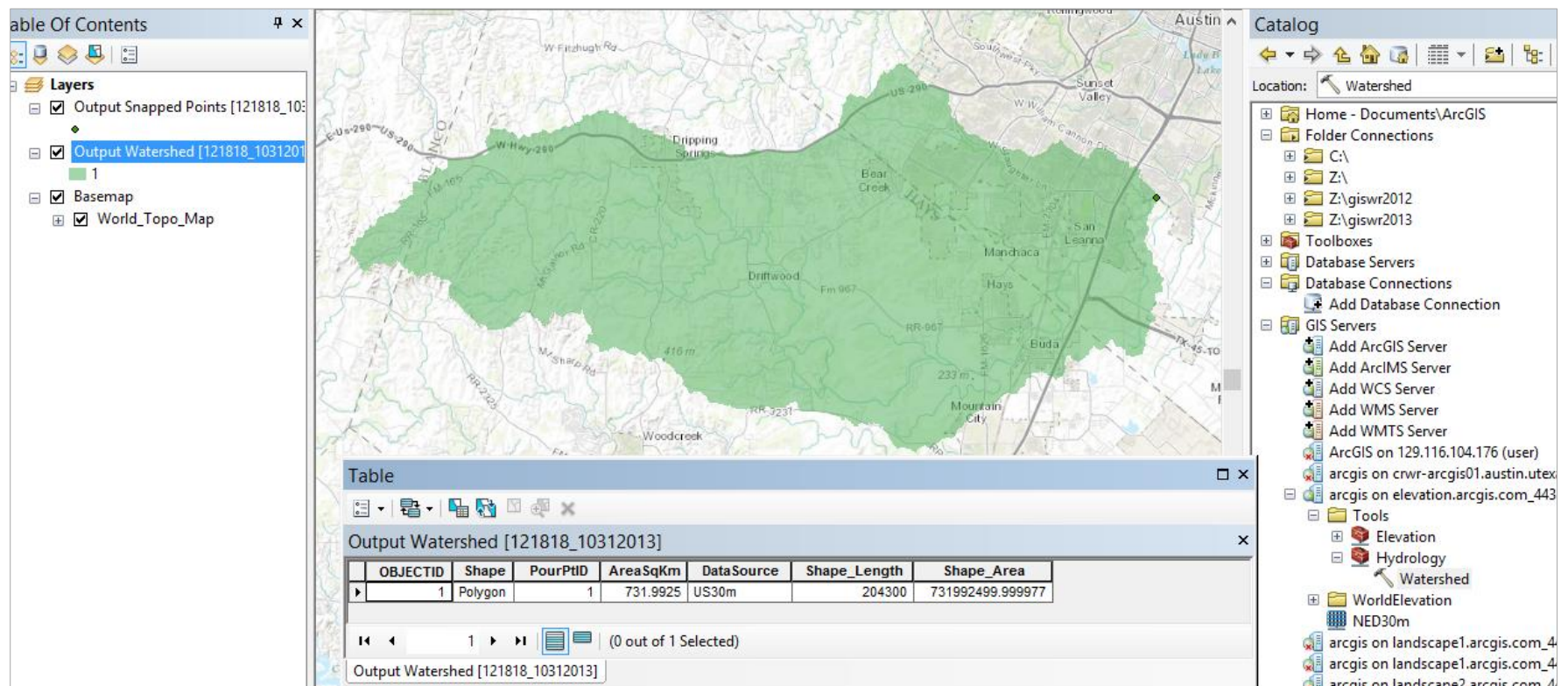


FEMA Flood Hazard Zone



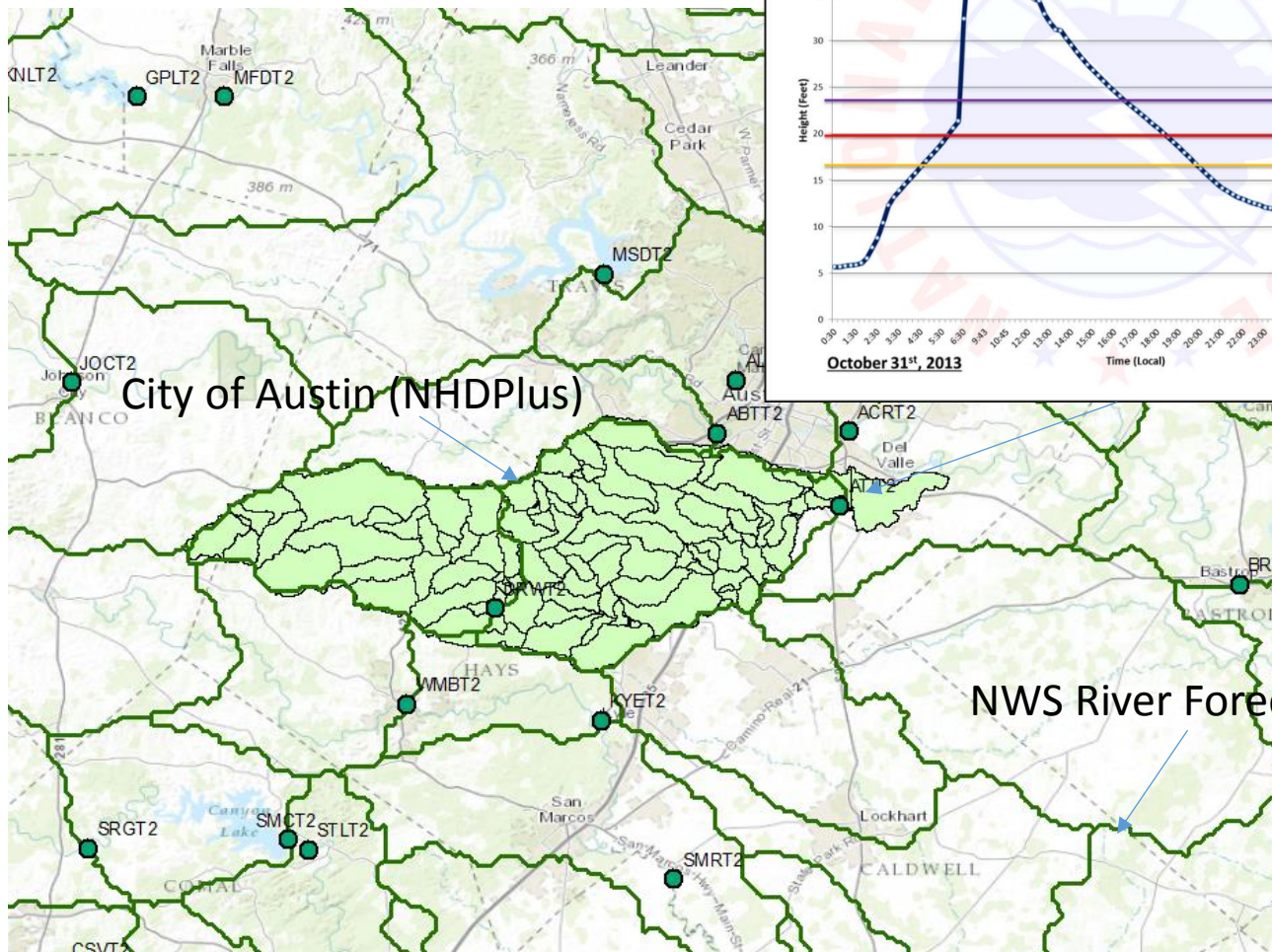
Drainage Area above Home

Drainage area = 732 km², or 282 miles²

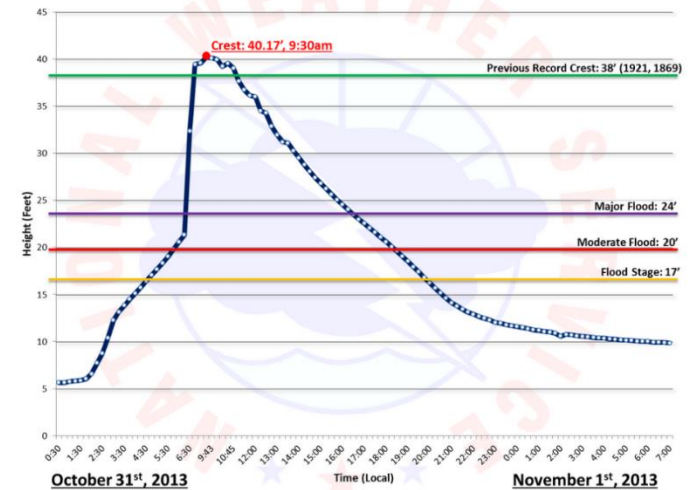


Watershed delineation using ESRI Watershed Delineation Services

National and Local Flood Modeling

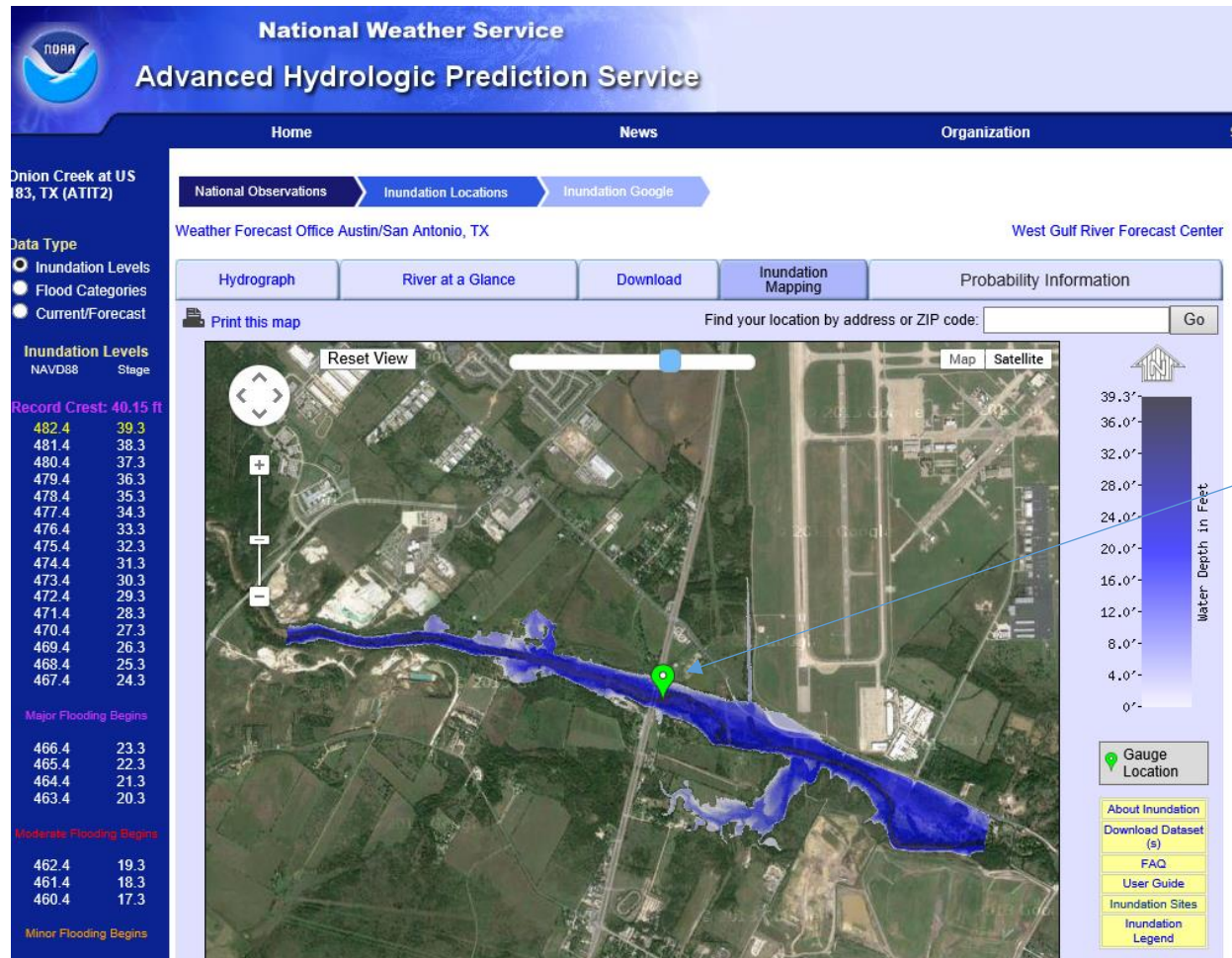


Onion Creek @ HWY 183 – ATIT2 – River Gauge



NWS River Forecast Center

Real-Time Flood Inundation Mapping on Onion Creek (USGS/NWS)



Gage at Highway 183

http://water.weather.gov/ahps2/inundation/inundation_google.php?gage=atit2

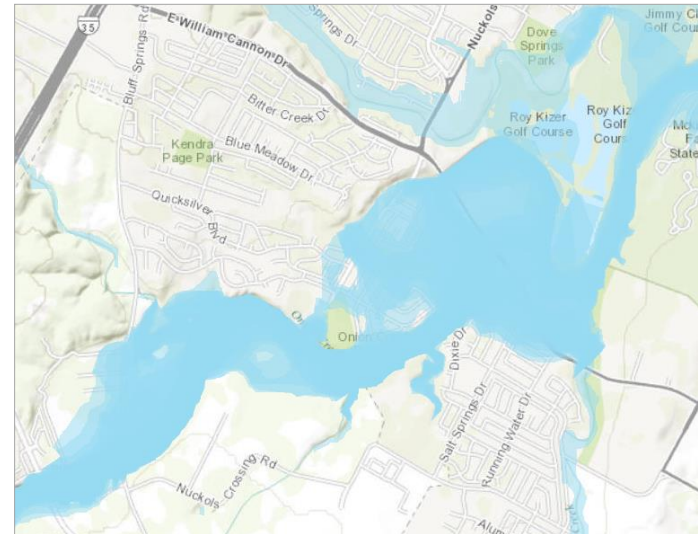
Mitigation and Response Levels

Mitigation (ft above NAVD88)

Response (Stage Height, ft)

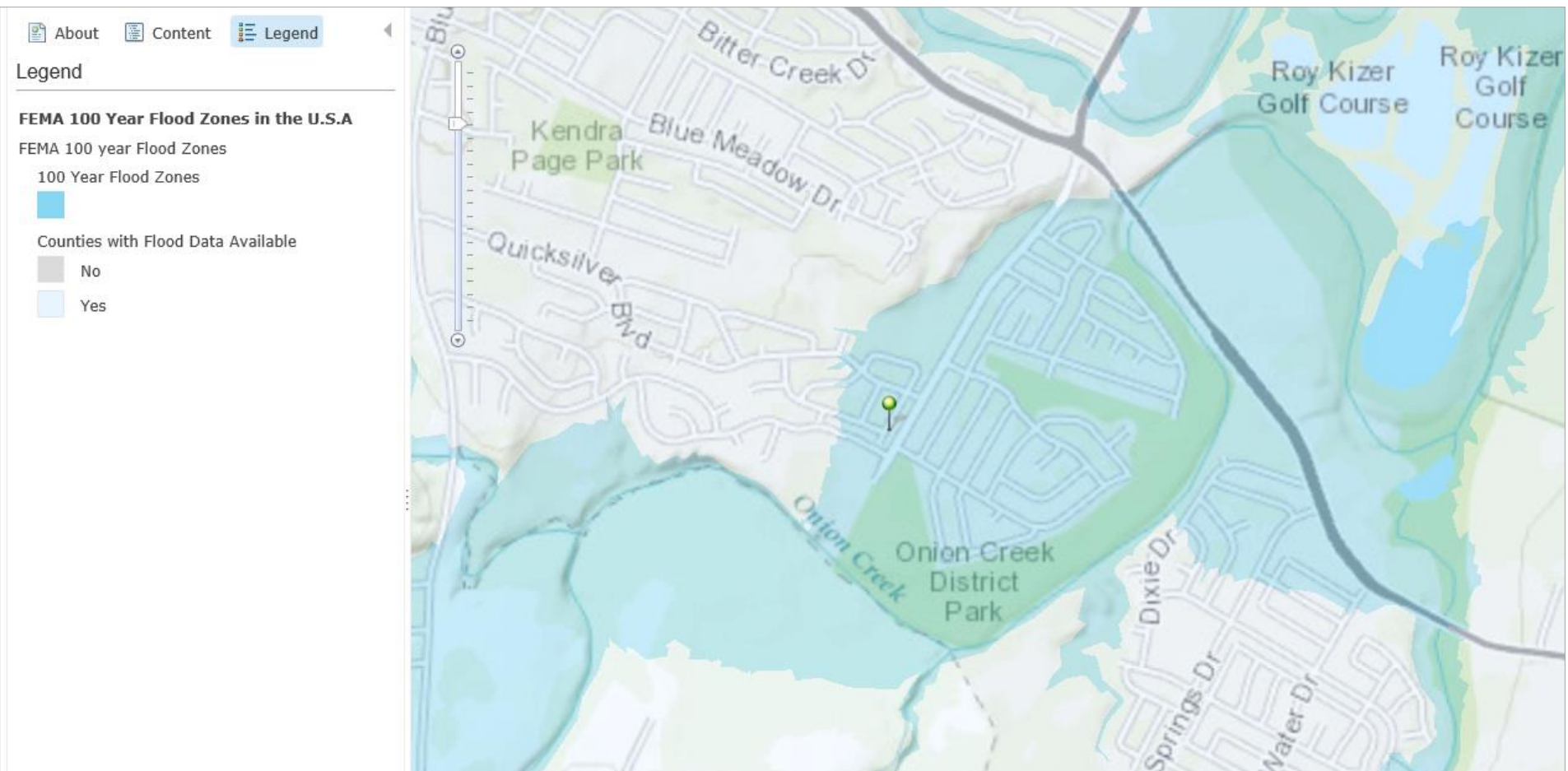
Onion Creek at US 183, TX (ATIT2)	
Data Type	
☒ Inundation Levels	
☐ Flood Categories	
☐ Current/Forecast	
Inundation Levels	
NAVD88	Stage
Record Crest: 40.15 ft	
482.4	39.3
481.4	38.3
480.4	37.3
479.4	36.3
478.4	35.3
477.4	34.3
476.4	33.3
475.4	32.3
474.4	31.3
473.4	30.3
472.4	29.3
471.4	28.3
470.4	27.3
469.4	26.3
468.4	25.3
467.4	24.3
Major Flooding Begins	
466.4	23.3
465.4	22.3
464.4	21.3
463.4	20.3
Moderate Flooding Begins	
462.4	19.3
461.4	18.3
460.4	17.3
Minor Flooding Begins	
459.4	16.3
458.4	15.3
* = Extended rating	

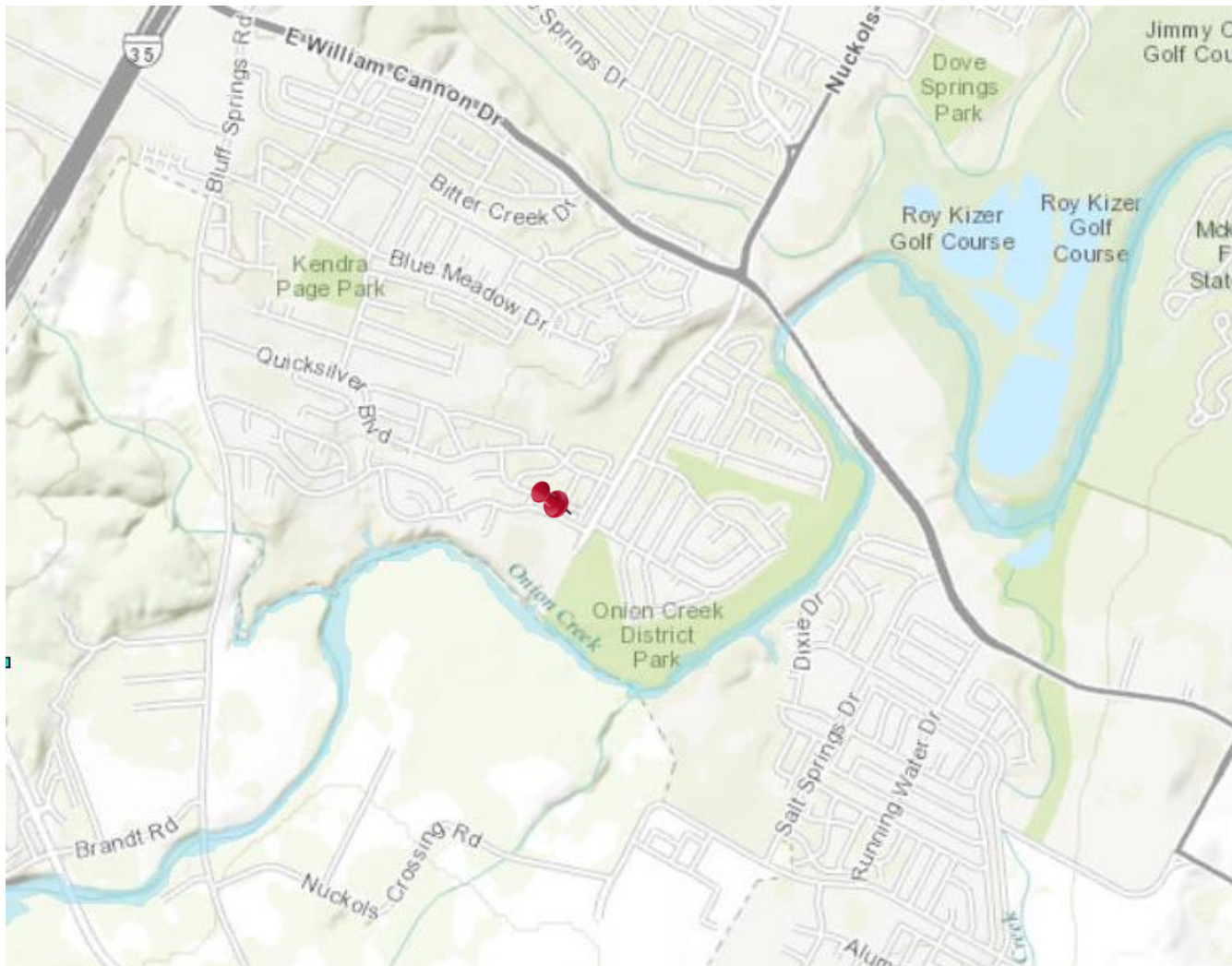
500 yr	490.34
200 yr	487.88
100 yr	484.82
50 yr	481.24
25 yr	477.09
10 yr	471.01
5 yr	466.62
2 yr	460.30



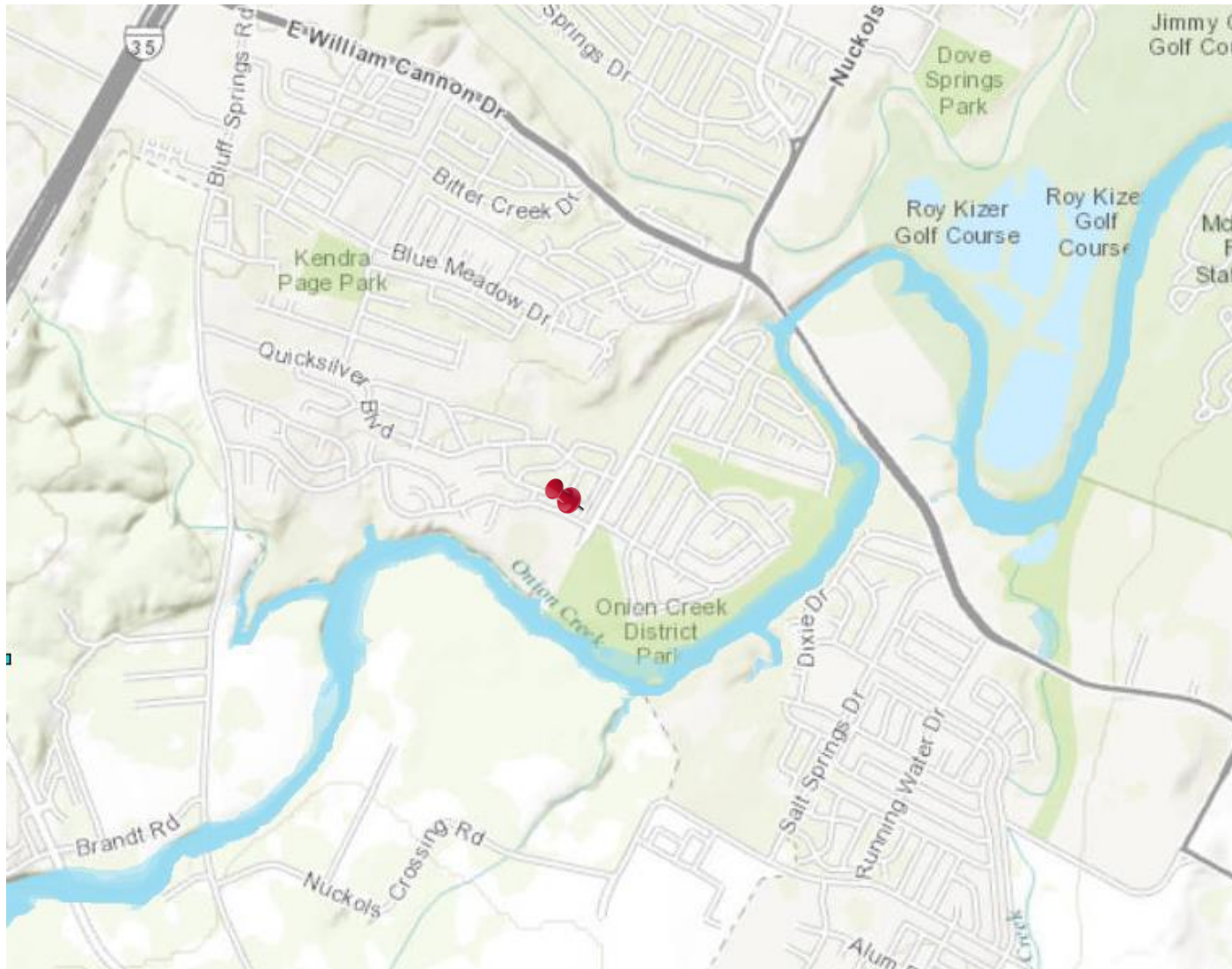
Major Flood Stage: 24
 Moderate Flood Stage: 20
 Flood Stage: 17
 Action Stage: 15

FEMA Flood Hazard Zone

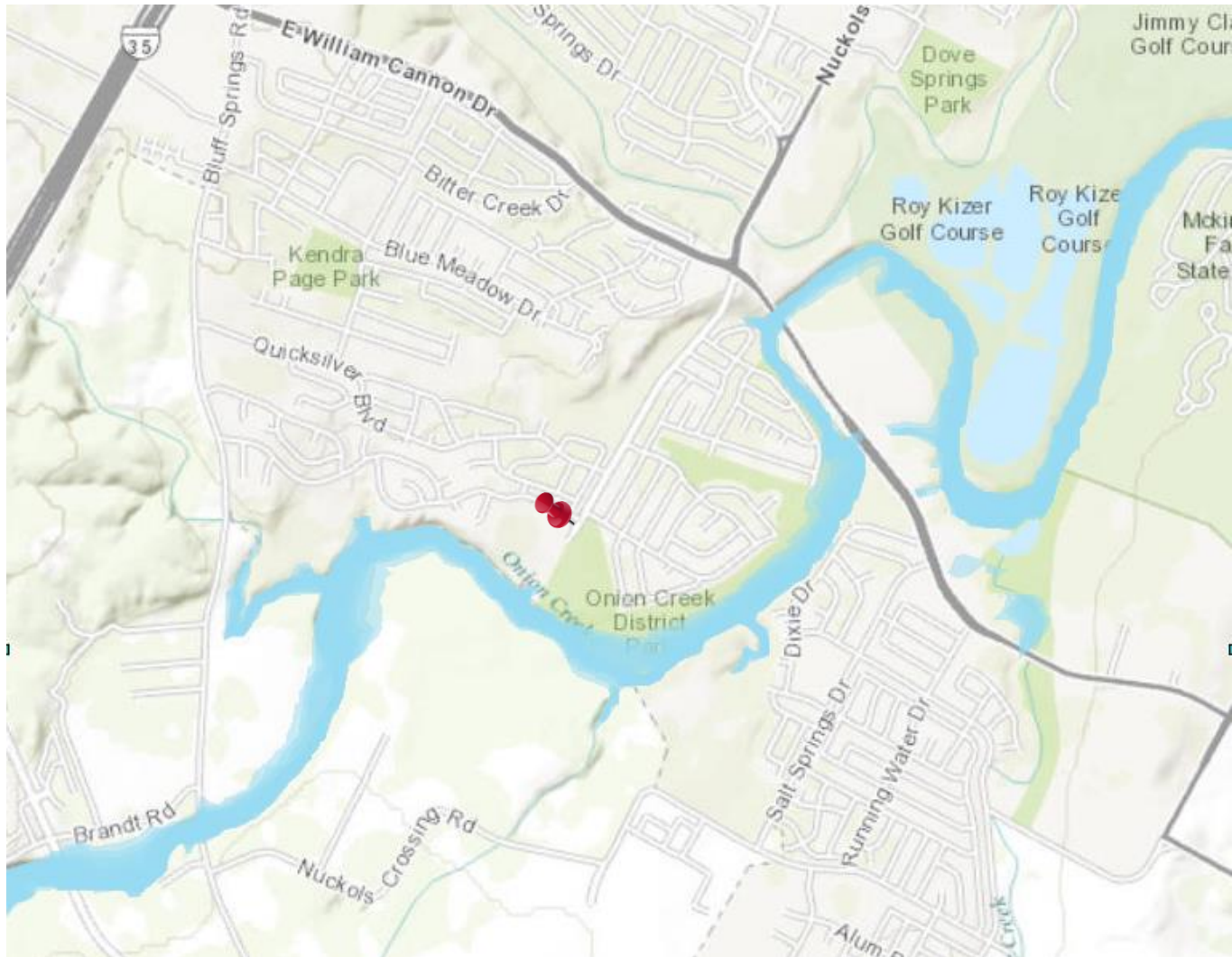




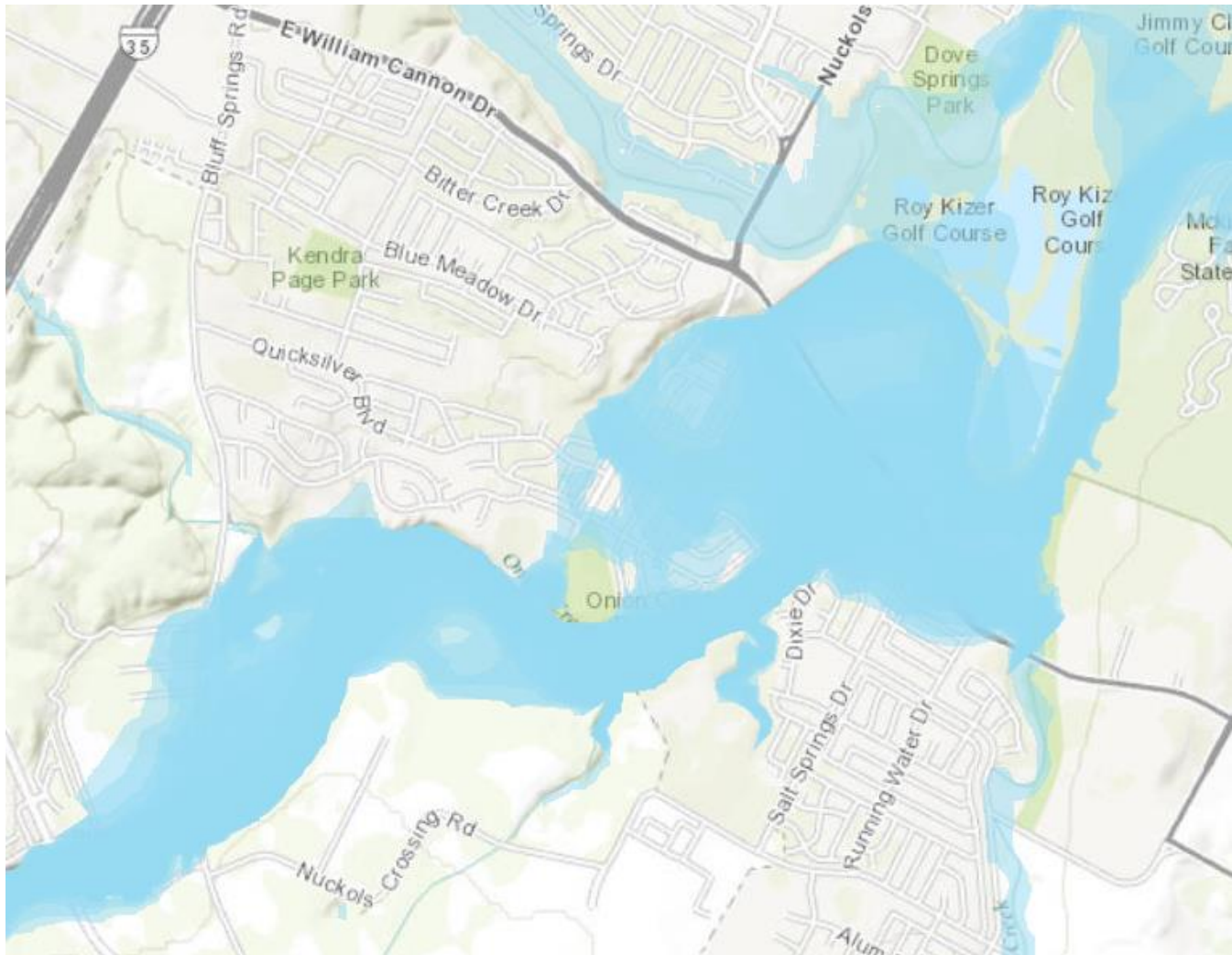
17' AHPS Flood
Warning Stage



21' NWS issues Flood Warning for Onion Creek
(issued at 02:44)

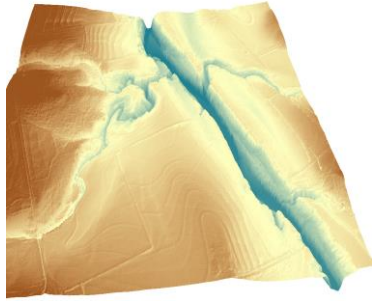


25' NWS issues Flood Warning for Onion Creek at 03:40 – gauge will rise to 25.9'

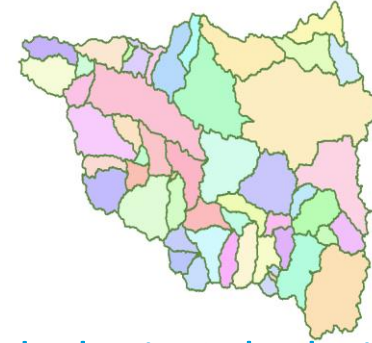


41' -- At 10:07 -- peak of USGS gauge -- 40.97'

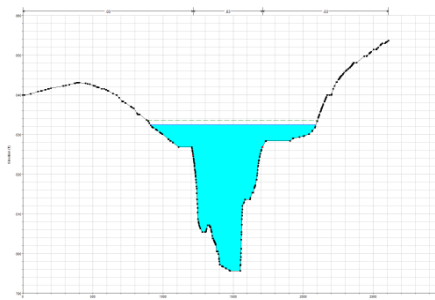
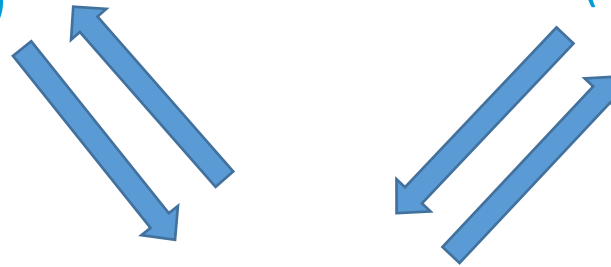
RiverML: Modular River Data Format



Terrain Processing Software
(ArcGIS, AutoCAD, etc.)

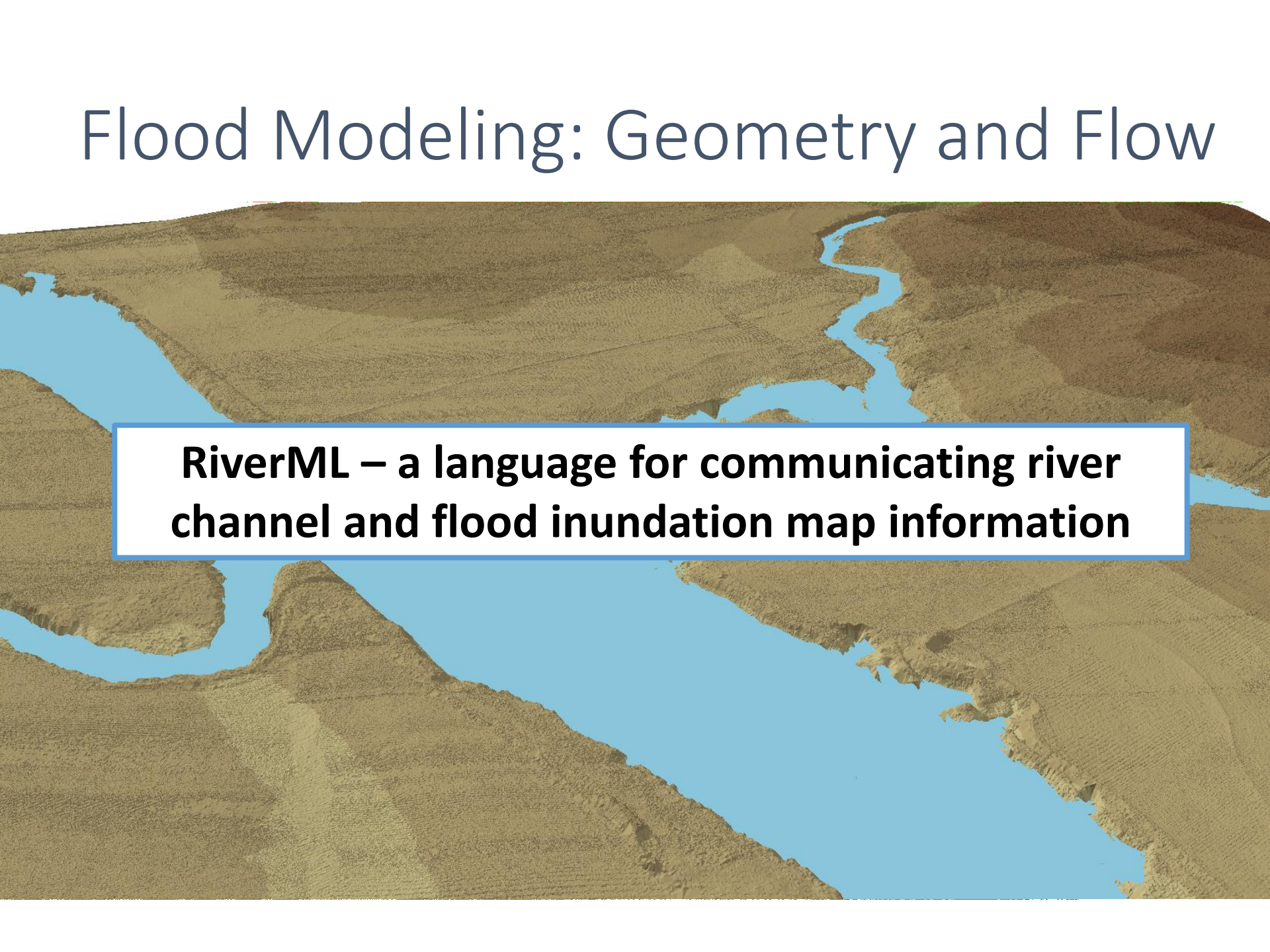


Hydrologic Calculation Software
(HEC-HMS, PondPack, etc.)



Hydraulic Calculation Software
(HEC-RAS, MIKE Flood HD, etc)

Flood Modeling: Geometry and Flow

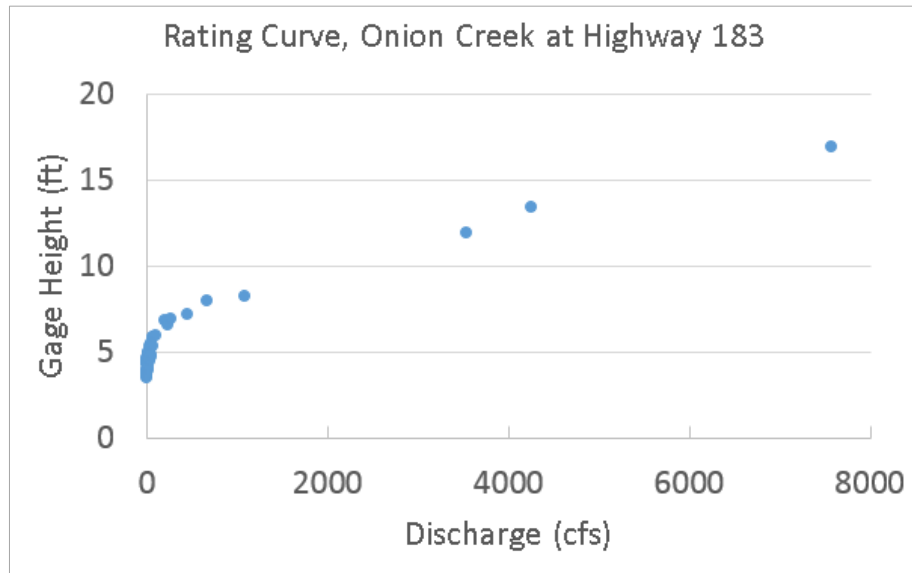


RiverML – a language for communicating river channel and flood inundation map information

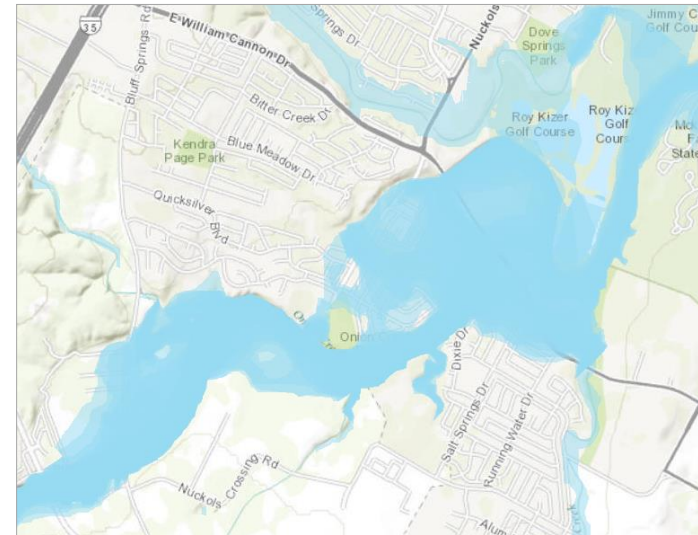
Rating Curve and Map

Can “Ratings and Gagings” work be extended to include an inundation map?

Rating Curve (Stage Height, ft)



Flood Inundation Map (Stage Height, ft)



Major Flood Stage: 24

Moderate Flood Stage: 20

Flood Stage: 17

Action Stage: 15

*Should stage height or
height above geodetic datum be used?*

Thoughts for HDWG

- The **Open Water Data Initiative** is starting up
 - larger scale engagement by the US federal agencies in water data exchange standards
- The **National Flood Interoperability Experiment** is happening
 - Significant engagement by the academic community
 - Educational materials and teaching programs about the data exchange architecture
 - Focus on flood information this year, especially forecasting and mapping
 - Reinvigorate Hydrometeorological Forecasting IE?